

Yield-Aware CMOS Inverter Layout Design and Defect Analysis Using KLayout

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Abstract

The objective of this project was to design a CMOS inverter layout using KLayout and investigate common manufacturing defects that affect semiconductor yield. A CMOS inverter was implemented using five fundamental fabrication layers: N-Well, Active, Polysilicon, Contact, and Metal1. After creating a defect-free layout, various yield-limiting defects including missing contacts, metal opens, metal shorts, spacing violations, and poly gate breaks were introduced and analyzed. The project demonstrates how fabrication defects can impact circuit functionality and manufacturing yield, thereby providing practical exposure to concepts relevant to Yield Engineering and Semiconductor Manufacturing.

1. Introduction

Semiconductor manufacturing involves numerous fabrication steps such as oxidation, lithography, implantation, etching, deposition, and metallization. Small process variations or fabrication defects can significantly impact device performance and manufacturing yield.

Yield Engineering focuses on identifying, analyzing, and minimizing such defects to improve the percentage of functional chips produced on a wafer.

In this project, a CMOS inverter was designed using KLayout and several manufacturing defects were intentionally introduced to understand their effect on circuit functionality.

2. Objectives

The primary objectives of this project were:

- To understand CMOS layout design methodology.
 - To gain practical experience with KLayout.
 - To study semiconductor fabrication layers.
 - To analyze common manufacturing defects.
 - To understand the relationship between defects and yield loss.
 - To develop skills relevant to Yield Engineering roles.
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3. Software Used

Software:

- KLayout 0.30.9

Operating System:

- Windows

Purpose:

- Layout design
 - Defect analysis
 - Layout visualization
 - Semiconductor mask representation
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4. CMOS Inverter Design

4.1 CMOS Inverter Structure

A CMOS inverter consists of:

- One PMOS transistor
- One NMOS transistor

Operation:

- Input LOW $\rightarrow$ PMOS ON, NMOS OFF $\rightarrow$ Output HIGH
- Input HIGH $\rightarrow$ PMOS OFF, NMOS ON $\rightarrow$ Output LOW

Thus the output is always the logical inversion of the input.

4.2 Layers Used

Layer 1: N-Well

Purpose:

- Formation of PMOS transistor

Dimensions:

- Width = 10 μm
 - Height = 5 μm
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Layer 2: Active Layer

Purpose:

- Source and drain diffusion regions

PMOS Active:

- Width = 8 μm
- Height = 1.5 μm

NMOS Active:

- Width = 8 μm
 - Height = 1.5 μm
-

Layer 3: Polysilicon

Purpose:

- Gate formation

The polysilicon gate intersects both PMOS and NMOS active regions, thereby forming the transistor channels.

Layer 4: Contact

Purpose:

- Electrical connection between diffusion regions and Metall

Total Contacts:

- PMOS Source Contact
- PMOS Drain Contact
- NMOS Source Contact
- NMOS Drain Contact

Total = 4 Contacts

Layer 5: Metall

Purpose:

- Routing and interconnection

Metall was used to implement:

- VDD
 - GND
 - Input Node (IN)
 - Output Node (OUT)
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5. Defect-Free Layout

A complete CMOS inverter layout was successfully implemented using all five layers.

The layout included:

- N-Well region
- PMOS transistor
- NMOS transistor
- Polysilicon gate
- Contacts
- Metal routing
- Input and output nodes

Figure 1

Insert screenshot of the final CMOS inverter layout.

6. Defect Analysis

Several common manufacturing defects were introduced into the layout to study their effect on circuit functionality.

6.1 Missing Contact Defect

Description

One PMOS drain contact was removed from the layout.

Manufacturing Causes

- Contact etch failure
- Misalignment
- Contact fill failure

Effect

The PMOS drain becomes electrically disconnected from the output node.

Result:

- Open circuit condition
- Functional failure

Figure 2

Insert Missing Contact Defect screenshot.

6.2 Metal Open Defect

Description

A discontinuity was introduced in the output metal path.

Manufacturing Causes

- Over-etching
- CMP defects
- Metal cracking

Effect

The output node becomes disconnected.

Result:

- Signal interruption
- Circuit failure

Figure 3

Insert Metal Open Defect screenshot.

6.3 Metal Short Defect

Description

An unintended metal bridge was introduced between VDD and OUT.

Manufacturing Causes

- Particle contamination
- Lithography bridging
- Residual metal after etching

Effect

Creates an unintended electrical connection.

Result:

- Excessive current flow
- Leakage
- Potential chip failure

Figure 4

Insert Metal Short Defect screenshot.

6.4 Spacing Violation

Description

Metal routing was intentionally placed closer than recommended spacing limits.

Manufacturing Causes

- Lithography resolution limits
- Overlay errors
- Process variation

Effect

Increased probability of bridging.

Result:

- Reduced yield
- Potential shorts

Figure 5

Insert Spacing Violation screenshot.

6.5 Poly Gate Break

Description

The polysilicon gate was intentionally broken into two disconnected sections.

Manufacturing Causes

- Poly etch defects
- Patterning failure
- Line break defects

Effect

Gate signal cannot propagate through the entire transistor structure.

Result:

- Transistor cannot switch correctly
- Functional failure

Figure 6

Insert Poly Gate Break screenshot.

7. Yield Engineering Perspective

Yield engineers analyze defects that occur during semiconductor manufacturing and identify their root causes.

The defects studied in this project represent real manufacturing challenges encountered in semiconductor fabrication facilities.

The project demonstrates how:

- Contact failures cause opens.
- Metal defects cause shorts and opens.
- Lithography issues cause spacing violations.
- Poly defects affect transistor operation.

Understanding these failure mechanisms is essential for improving process yield and manufacturing reliability.

8. Skills Acquired

Through this project, the following skills were developed:

- CMOS layout design
- Semiconductor fabrication understanding
- KLayout usage
- Defect analysis
- Yield engineering concepts
- Failure mechanism analysis
- Semiconductor manufacturing awareness

9. Future Work

The project can be extended by:

1. Designing NAND and NOR gate layouts.
2. Performing Design Rule Check (DRC).
3. Implementing SRAM cell layouts.
4. Studying critical area analysis.
5. Simulating CMOS inverter functionality in LTspice.
6. Performing Monte Carlo variation analysis.
7. Developing a yield-aware standard cell library.

10. Conclusion

A CMOS inverter was successfully designed using KLayout by implementing five essential fabrication layers: N-Well, Active, Polysilicon, Contact, and Metal1. Multiple manufacturing defects including missing contacts, metal opens, metal shorts, spacing violations, and poly gate breaks were introduced and analyzed. The study demonstrated how fabrication defects directly affect circuit functionality and manufacturing yield. The project provided practical exposure to semiconductor layout design and yield engineering concepts relevant to modern semiconductor manufacturing industries.
